

Project Proposal
CIS 602-7103: Special Topics in CIS

**Topic: Analysing Movie Reviews with Sentiment Analysis
Using AWS**

Group:15

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1. Introduction

The project aims to analyse movie reviews from a dataset to determine the sentiment (positive, negative, or neutral) using natural language processing (NLP). This project will utilize AWS services for data storage, processing, and visualization.

2. Objectives

- To collect and preprocess movie reviews data.
- To perform sentiment analysis on the reviews.
- To store and process data using AWS services.
- To visualize the sentiment analysis results.

3. Scope

- Data collection and preprocessing.
- Sentiment analysis using NLP.
- Utilizing AWS for data storage, processing, and visualization.
- Creating a dashboard to display results.

4. Methodology

4.1 Data Collection

- **Dataset:** Use an existing movie reviews dataset (e.g., from Kaggle).
- **Storage:** Upload the dataset to AWS S3 for centralized storage.

4.2 Data Preprocessing

- **Cleaning:** Remove any irrelevant information, handle missing values, and preprocess text (e.g., tokenization, stop words removal).
- **AWS Service:** Use AWS Lambda for preprocessing the data.

4.3 Sentiment Analysis

- **NLP:** Apply sentiment analysis using Python libraries such as NLTK or TextBlob.
- **AWS Service:** Use AWS SageMaker for training and deploying the sentiment analysis model.

4.4 Data Processing

- **Batch Processing:** Use AWS Glue for ETL (Extract, Transform, Load) operations on the processed data.

4.5 Data Visualization

- **Dashboard:** Create a dashboard to visualize sentiment analysis results.
- **AWS Service:** Use AWS Quick Sight for creating and managing the dashboard.

5. Technologies and Tools

- **Data Storage:** AWS S3
- **Data Preprocessing:** AWS Lambda, Python
- **Sentiment Analysis:** AWS SageMaker, Python (NLTK, TextBlob)
- **Data Processing:** AWS Glue
- **Data Visualization:** AWS Quick Sight

6. Expected Outcomes

- A cleaned and pre-processed dataset of movie reviews.
- A trained sentiment analysis model deployed on AWS SageMaker.
- A processed and analysed dataset with sentiment scores.
- A dashboard visualizing the sentiment analysis results.

7. Risk Management

- **Data Privacy:** Ensure data privacy by anonymizing any personal information in the dataset.
- **System Reliability:** Implement error handling and logging in AWS Lambda functions.
- **Scalability:** Design the system to handle larger datasets if needed.

Conclusion

This project will provide hands-on experience with AWS services and big data technologies. The sentiment analysis of movie reviews will demonstrate the power of cloud-based data processing and visualization, offering valuable insights into public opinion about movies.